

eg-1] The table gives the distance in nautical miles of the visible horizon for the given heights in feet about the earth's surface by using N.F.M.

$x =$ Height	100	150	200	250	300	350	400
$y =$ distance	10.63	13.03	15.04	16.81	18.42	19.90	21.27

find the value of y when; $f(x) = f(130)$.

→ Ans-

x	$f(x)$	Δy_0	$\Delta^2 y_0$	$\Delta^3 y_0$	$\Delta^4 y_0$	$\Delta^5 y_0$	$\Delta^6 y_0$
100	10.63						
		-2.4					
150	13.03		-0.39				
		2.01		+0.15			
200	15.04		-0.24		-0.07		
		1.77		0.08		+0.028	
250	16.81		-0.16		-0.05		0.02
		1.61		0.03		-0.04	
300	18.42		-0.13		-0.01		
		1.48		0.02			
350	19.90		-0.11				
		1.37					
400	21.27						

we have,

$$p = \frac{x - x_0}{h}$$

∴ Here, $x = 130$; $x_0 = 100$; $h = 50$

$$\therefore p = \frac{130 - 100}{50} \therefore p = 0.6$$

Then, we have,

$$\begin{aligned} \gamma &= f(x) + p \Delta \gamma_0 + \frac{p(p-1)}{2!} \Delta^2 \gamma_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 \gamma_0 \\ &+ \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 \gamma_0 + \frac{p(p-1)(p-2)(p-3)(p-4)}{5!} \Delta^5 \gamma_0 \\ &+ \frac{p(p-1)(p-2)(p-3)(p-4)(p-5)}{6!} \Delta^6 \gamma_0 \end{aligned}$$

$$\therefore \gamma = 10.63 + (0.6 \times 2.4) + \frac{0.6(0.6-1)}{(1 \times 2)} x(-0.39) +$$

$$\frac{0.6(0.6-1)(0.6-2)}{(3 \times 2 \times 1)} x 0.15 x 0.15^2 +$$

$$\frac{0.6(0.6-1)(0.6-2)(0.6-3)(-0.07)}{(4 \times 3 \times 2 \times 1)} +$$

$$\frac{0.6(0.6-1)(0.6-2)(0.6-3)(0.6-4)}{(5 \times 4 \times 3 \times 2 \times 1)} x (0.02) +$$

$$\frac{0.6(0.6-1)(0.6-2)(0.6-3)(0.6-4)(0.6-5)}{(6 \times 5 \times 4 \times 3 \times 2 \times 1)} x (0.02)$$

$$\therefore \gamma = 10.63 + 1.44 + 0.0468 + 8.4 \times 10^{-3} + 2.352 \times 10^{-3} +$$

$$4.5696 \times 10^{-4} - 3.351 \times 10^{-4}$$

$$\therefore \underline{\underline{\gamma = 12.1277}}$$

eg-1 find $f(42)$ for the following data :

x	20	25	30	35	40	45
$f(x)$	354	332	291	260	231	204

using Newton's backward diff. method.

→ Ans.

x	$f(x)$	∇y_n	$\nabla^2 y_n$	$\nabla^3 y_n$	$\nabla^4 y_n$	$\nabla^5 y_n$
20	354					
		-22				
25	332		-19			
		-41		29		
30	291		10		-37	
		-31		-8		-45
35	260		2		8	
		-29		0		
40	231		2			
		-27				
45	204					

formula-

$$\begin{aligned}
 y = & y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n \\
 & + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_n + \frac{p(p+1)(p+2)(p+3)(p+4)}{5!} \nabla^5 y_n
 \end{aligned}$$

$$\therefore y =$$

$$\therefore p = \frac{a - 2n}{h}$$

$$\therefore p = \frac{42 - 40}{5}$$

$$\therefore p = 0.4$$

$$\therefore y = 231 + 0.4 \times (-29) + \frac{0.4 \times (0.4 + 1) \times 2}{(1 \times 2)} +$$

$$\frac{0.4 \times (0.4 + 1) \times (0.4 + 2) \times (-8)}{(1 \times 2 \times 3)} +$$

$$\frac{0.4 \times (0.4 + 1) \times (0.4 + 2) \times (0.4 + 3) \times (-31)}{(1 \times 2 \times 3 \times 4)} +$$

$$\frac{0.4 \times (0.4 + 1) \times (0.4 + 2) \times (0.4 + 3) \times (0.4 + 4) \times (-4)}{(1 \times 2 \times 3 \times 4 \times 5)}$$

$$\therefore y = 231 - 11.6 + 0.56 - 1.792 - 7.05 = 21.118$$

$$\therefore y = 21.118$$

eg 3] find $f(17)$ by using N. B. D. M.

x	0	5	10	15	20
y	1.0	1.6	3.8	8.2	15.4

→ Ans -

x	$f(x)$	∇y_n	$\nabla^2 y_n$	$\nabla^3 y_n$	$\nabla^4 y_n$
0	1.0				
		0.6			
5	3.8 1.6		1.6		
		2.2		0.6	
10	8.2 3.8		2.2		0
		4.4		0.6 = 0	
15	8.2		2.8		
		7.2			
20	15.4				

formula -

$$y = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n$$

$$+ \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_n$$

where, $p = \frac{x - x_n}{h}$ $\therefore p = \frac{17 - 15}{5} \therefore p = 0.4$

$$\therefore y = 8.2 + 0.4 \times 4.4 + 0.4(0.4+1) \cdot 2.2 + 0.4(0.4+1)(0.4+2) \cdot 0.6 + \frac{(1 \times 2 \times 3)}{4!} \cdot 0$$

$$\frac{0.4(0.4+1)(0.4+2)(0.4+3)}{(1 \times 2 \times 3 \times 4)} \cdot 0$$

$$\therefore y = 8.2 + 1.76 + 0.62 + 0.134 + 0$$

$$\therefore \underline{\underline{y = 10.714}}$$

Eg-4] find the polynomial $f(x)$ by using Lagrange's formula & hence find $f(3)$ for -

x	0	1	2	5
$f(x)$	2	3	12	147

→ Ans -

Here, $n=3$

$$x_0=0 ; x_1=1 ; x_2=2 ; x_3=5$$

$$y_0=2 ; y_1=3 ; y_2=12 ; y_3=147$$

Then,

formula -

$$y = \frac{(x-x_1)(x-x_2)(x-x_3) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2)(x_0-x_3) \dots (x_0-x_n)} x y_0 +$$

$$\frac{(x-x_0)(x-x_2)(x-x_3) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2)(x_1-x_3) \dots (x_1-x_n)} x y_1 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_3) \dots (x-x_n)}{(x_2-x_0)(x_2-x_1)(x_2-x_3) \dots (x_2-x_n)} x y_2 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_2) \dots (x-x_n)}{(x_3-x_0)(x_3-x_1)(x_3-x_2) \dots (x_3-x_n)} x y_3 +$$

$$\therefore \gamma = \frac{(x-1)(x-2)(x-5)}{(0-1)(0-2)(0-5)} x^2 +$$

$$\frac{(x-0)(x-2)(x-5)}{(1-0)(1-2)(1-5)} x^3 + \frac{(x-0)(x-1)(x-5)}{(2-0)(2-1)(2-5)} x^1 +$$

$$\frac{(x-0)(x-1)(x-2)}{(5-0)(5-1)(5-2)} x^{147}.$$

$$\therefore \gamma = \left(\frac{2x^3 - 16x^2 + 34x - 20}{-10} \right) + \left(\frac{3x^3 - 21x^2 + 30x}{4} \right)$$

$$- \left(\frac{12x^3 - 60x^2 - 12x^2 + 60x}{60} \right) + \left(\frac{147x^3 - 147x^2 + 294x}{80} \right)$$

$$\therefore \gamma = - \left(\frac{12x^3 - 96x^2 + 204x - 120}{60} \right) + \left(\frac{45x^3 - 315x^2 + 450x}{60} \right)$$

$$- \left(\frac{120x^3 - 600x^2 - 120x^2 + 600x}{60} \right) + \left(\frac{147x^3 - 294x^2 + 147x^2 + 294x}{60} \right)$$

$$\therefore \gamma = 192x^3 - 519x^2 + 57x^2 + 744x - 132x^3 + 376x^2 - 804x + 120$$

$$\therefore \gamma = 60x^3 + 60x^2 - 60x + 120$$

$$\therefore \gamma = x^3 + x^2 - x + 2$$

$$\therefore \gamma = x^3 + x^2 - x + 2$$

Eg-5] Find the missing term in the following table using interpolation.

x	0	1	2	3	4
y	1	3	9	---	81

→ Ans-

Here,

$$x_0 = 0 ; x_1 = 1 ; x_2 = 2 ; x_3 = 4$$

$$y_0 = 1 ; y_1 = 3 ; y_2 = 9 ; y_3 = 81$$

$$x = 3 \quad (\because f(3) = f(x))$$

$$y = \frac{(x-x_1)(x-x_2)(x-x_3) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2)(x_0-x_3) \dots (x_0-x_n)} \times y_0 +$$

$$\frac{(x-x_0)(x-x_2)(x-x_3) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2)(x_1-x_3) \dots (x_1-x_n)} \times y_1 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_3) \dots (x-x_n)}{(x_2-x_0)(x_2-x_1)(x_2-x_3) \dots (x_2-x_n)} \times y_2 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_2) \dots (x-x_n)}{(x_3-x_0)(x_3-x_1)(x_3-x_2) \dots (x_3-x_n)} \times y_3$$

$$\therefore y = \frac{(2)(1)(-1)}{(-1)(-2)(-4)} \times 1 + \frac{(3)(1)(-1)}{(1)(-1)(-3)} \times 3 +$$

$$\frac{(3)(2)(-1)}{(2)(1)(-2)} \times 9 + \frac{(3)(2)(1)}{(4)(3)(2)} \times 81$$

$$y = 0.25 - 3 + 13(5 + 20.25)(1) + 271 - 1.4$$

$$\therefore y = 31$$

eg-7 Given the values -

x	5	7	11	13	17
$f(x)$	150	392	1452	2366	5202

Evaluate $f(9)$; Using Newton's divided diff. method.

Ans -

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
5	150				
7	392	$392 - 150 = 242$			
11	1452	$1452 - 392 = 1060$	$1060 - 242 = 818$		
13	2366	$2366 - 1452 = 914$	$914 - 1060 = -146$	$-146 - 818 = -964$	
17	5202	$5202 - 2366 = 2836$	$2836 - 914 = 1922$	$1922 - (-964) = 2886$	$2886 - (-964) = 3850$

where, $A(x-x_0)(x-x_1) + \dots + A(x-x_0)(x-x_1)(x-x_2)(x-x_3)(x-x_4)$

$$y = y_0 + (x-x_0) \Delta y_0 + (x-x_0)(x-x_1) \Delta^2 y_0 +$$

$$(x-x_0)(x-x_1)(x-x_2) \Delta^3 y_0 + (x-x_0)(x-x_1)(x-x_2)(x-x_3) \Delta^4 y_0$$

$$\therefore \gamma = 150 + (4)(121) + (4)(2)(24) + (4)(2)(-2)(1)$$

$$\therefore \gamma = 150 + 484 + 192 - 16.$$

$$\therefore \underline{\underline{\gamma = 810}}$$

Q] Using Euler's Method find the root of eqⁿ $\frac{dy}{dx} = -2x^3 + 12x^2 - 20x + 8.5$

from $x=0$ to $x=4$ with a step size of 0.5. The initial condition at $x=0$ is $y=1$.

Ans -

Given -

$\frac{dy}{dx} = -2x^3 + 12x^2 - 20x + 8.5$ at initial condition $x=0$ & $y=1$.

we take $n=9$ & $h=0.5$ which is sufficiently small.

The calculations are arranged as follows -

x	y	$\frac{dy}{dx} = -2x^3 + 12x^2 - 20x + 8.5$	New $y = \text{old } y + 0.5 \left(\frac{dy}{dx} \right)$
0	1	8.5	5.25
0.5	5.25	7.25	5.875
1	5.875	-1.525	5.125
1.5	5.125	-1.25	4.5
2	4.5	0.5	4.75
2.5	4.75	2.25	5.875
3	5.875	2.5	7.125
3.5	7.125	-0.25	7.0
4	7		

$\therefore$ The value of y (Approx) is 7 at $x=4$.

Q. II] Using modified Euler's method, find $y(0.2)$ & $y(0.4)$ given $y' = y + e^x$, $y(0) = 0$

Ans -

we have, $y' = y + e^x$; $y(0) = 0$ & $h = 0.2$

① To calculate $y(0.2) \rightarrow$

x	$y + e^x = y'$	Mean slope	old $y + h(\text{mean slope}) = \text{new } y$
0.0	1	-	0.2
0.2	1.4214	1.2107	0.2421
0.2	1.4635	1.2317	0.2463
0.2	1.4617	1.2338	0.2468
0.2	1.4682	1.2341	0.2464

$\therefore y(0.2) = 0.2468$

② To calculate $y(0.4) \rightarrow$

x	$y + e^x = y'$	Mean slope	old $y + 0.2(\text{M.S}) = \text{new } y$
0.2	1.4682	-	0.5404
0.4	2.0322	1.7502	0.5968
0.4	2.0887	1.7784	0.6025
0.4	2.0943	1.78125	0.6030
0.4	2.0949	1.7815	0.6031
0.4	2.0949	1.7816	0.6031

$\therefore y(0.4) = 0.6031$

Q.12] Apply Runge-Kutta fourth order method to find an approximate value of y when $x=0.2$ given that $\frac{dy}{dx} = x+y$ & $y=1$ when $x=0$.

Ans. — Here, $x_0=0$, $y_0=1$, $h=0.2$; $f(x_0, y_0)=1$

$$k_1 = hf(x_0, y_0) = 0.2 \times 1 = 0.2000$$

$$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right) = 0.2 \times f(0.1, 1.1) = 0.2400$$

$$k_3 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right) = 0.2 \times f(0.1, 1.12) = 0.2440$$

$$k_4 = hf(x_0 + h, y_0 + k_3) = 0.2 \times f(0.2, 1.244) = 0.2888$$

$$y = y_0 + \frac{h}{6}(k_1 + 4k_2 + 2k_3 + k_4) = 1 + \frac{0.2}{6}(0.2000 + 4 \times 0.2400 + 2 \times 0.2440 + 0.2888) = 1.4568$$

$$\text{Hence, } y = 0.2428$$